A MatCont Tutorial
Computing Lyapunov Exponents

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1 Introduction

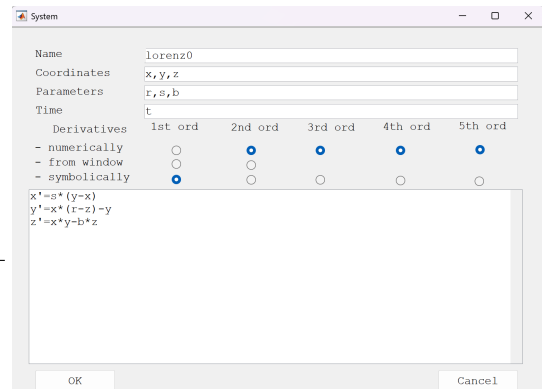
This tutorial explains how to compute Lyapunov Exponents (LEs) using the MatCont GUI [1]. This method has been introduced into MatCont with version 7p5. While most methods in MatCont compute invariant objects, it is hard to characterize chaotic dynamics but Lyapunov exponents may provide numerical indication of chaos.

The implementation employs the classical QR-algorithm, as provided by Govorukhin on Mathworks Central (with permission) [2]. For this QR-method, one integrates the variational equation $Y' = DF(x(t))Y$ with initial condition $Y(0) = I_n$ along an orbit $x(t)$ for a system $x' = F(x, p)$. After fixed intervals, one collects the growth factors and re-orthogonalizes the Lyapunov vectors represented by the columns of Y. These growth factors are used to compute all LEs. Computing LEs is typically done with separate code. However, once a user has entered a system in MatCont, the numerical exploration has direct access to the Jacobian matrix DF. Therefore, the numerical integration for an orbit can be augmented with the variational equation. So instead of separate codes, it may be convenient to have a method to compute LEs within MatCont.

In MatCont, one first chooses the type of an initial point. This could be some arbitrary point, an equilibrium or a bifurcation point. For LEs, we just choose some point labelled by P. Next, the user specifies the type of a solution curve. From an arbitrary point, one typically computes an orbit (O). We have now added a curve type Lyapunov Exponent (LE) that calls this method. We provide two options. First, for fixed values of the parameters, we report the running estimate of Lyapunov Exponents. This option allows to check numerical settings to inspect the convergence of LEs. Second, we allow to vary a single parameter for a specified array of values. This option allows to see how LEs of an attractor evolve as the parameter changes and also see bifurcations and the transition to chaos. One can also call the method from the command line, which still minimizes user effort but giving more control.

2 Computing Lyapunov exponents

We assume the first tutorial `lab1` has been followed. This tutorial explains the setup of MatCont, how to enter a system, and perform simulations (time integration). From there, we proceed to compute Lyapunov Exponents for the classical Lorenz system. We first enter the Lorenz system via Select ⟩ System ⟩ New and a pop-up window appears where we can specify the equations. As `lorenz` is an internal matlab command, we add another symbol to the name. We enter the following fields, see Table 1.



Name	Lorenz0
Coordinates	x,y,z
Parameters	r,s,b
Derivatives	Select first-order to be symbolic if possible.
System definition field	x'=s*(y-x) y'=x*(r-z)-y z'=x*y-b*z

Table 1: Specifying the Lorenz system. You can only select symbolic derivatives if you have the symbolic toolbox installed. In this tutorial, we do not need higher-order derivatives. Keeping "numerically" affects numerical results a little bit, but not much.

Now, we first want to compute all three Lyapunov Exponents for the classical parameters $r = 28, s = 10, b = 8/3$. we set the Point Type Type ⟩ Initial Point ⟩ Point and the Curve Type Type>Curve>Lyapunov Exponent. In the Starter and Integrator window, we specify several settings. We provide a nonzero initial condition $x = 0.1, y = 0, z = 0$ and set the parameters. We change the interval from 1 to 1000. Your windows should like in 1.

We are all set to compute the Lyapunov exponents via Compute ⟩ Forward. This takes a while, and our implementation does not allow for intermediate output as that is slowing down the computation too much. After a few seconds, the computation is ready, and we can inspect the results. We want to see all LEs as they are computed over time. Open a plot window via Window/Output ⟩ Graphic ⟩ 2D Plot and adjust the layout in the plot MatCont ⟩ Layout and select time and all LEs and adjust the ranges, see Figure 2. We see that the exponents tend to the known values $l_1 = 0.89, l_2 = 0.0005 \approx 0, l_3 = -14.50$.

We now show the other mode where a parameter can change. We declare the parameter r to be active by clicking the radio button. We want to vary r in steps of 0.05 around the subcritical Hopf bifurcation at $r = 24.74$ to observe the jump towards chaos. In the field *Active Parameter Range*, we set `24:.05:25`. This field could be any valid Matlab array. Note that these values are used for r and that 28 is not. Finally, we want to use the end state of the current computation as the initial condition for the next parameter value, and we check the box *Restart from last state*. The Starter window should now look as Figure 3(left).

Make sure to close all Plots and Numeric windows as otherwise the computation will take much longer. We press Compute ⟩ Forward and wait about a minute. We open the 2Dplot again, and adjust the layout so that the horizontal axis shows the parameter r. The ranges are adjusted to $[24, 25] \times [-0.1, 1.0]$. With 2DPlot:MatCont ⟩ Redraw Curve we then get the plot in 3(right).

In the mode that we vary parameters and follow an attractor, we may observe that an exponent increases towards zero and then decreases again or jumps. This typically indicates a bifurcation, and here Lyapunov Exponents provide a first scan for interesting parameter regimes. Indeed, before the subcritical Hopf bifurcation we have all exponents negative implying we follow an equilibrium and next the appearance

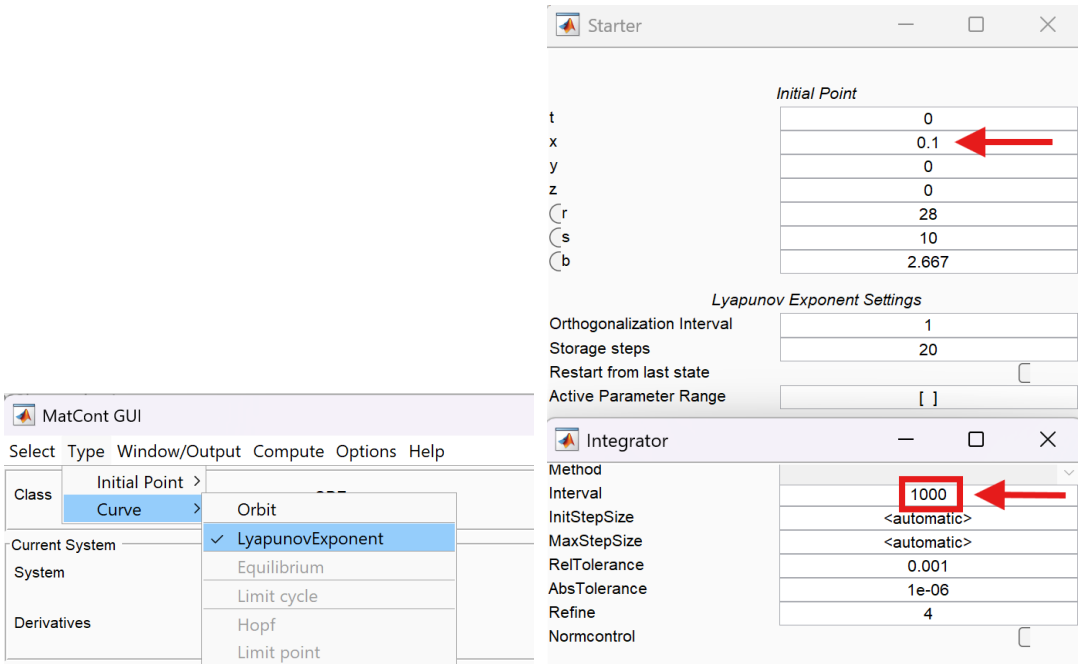


Figure 1: Initialization of the computation and numerical settings

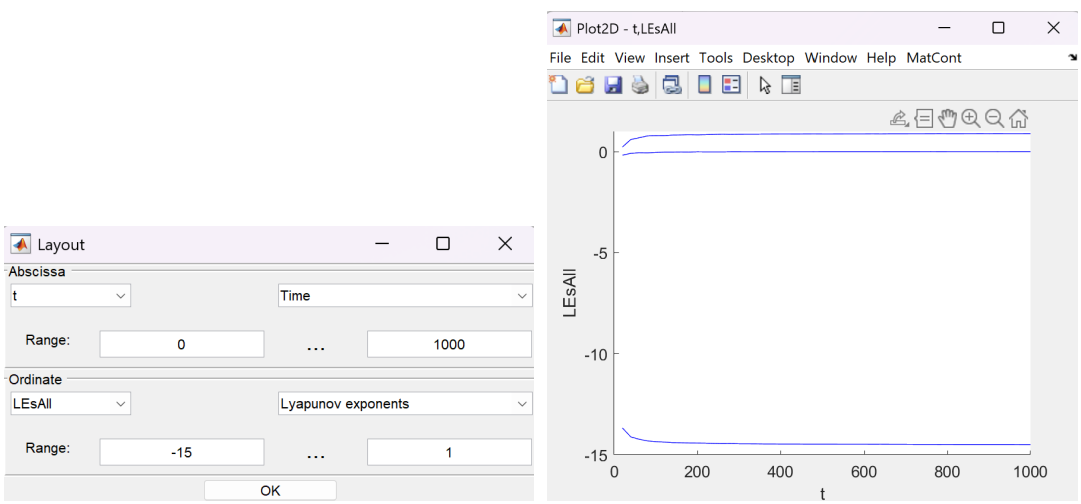


Figure 2: The running estimates of all Lyapunov exponents.

of a positive Lyapunov exponent near $r = 24.8$ shows the computation has jumped to the chaotic attractor.

2.1 Guidelines for algorithmic settings

- Since there is expansion and contraction, numerical errors enter and the vectors may get aligned. To prevent numerical artefacts, the Lyapunov vectors have to be orthogonalized. The field *Orthogonalization Interval* T specifies when this orthogonalization is done. We want $\exp(T(LE_{largest} - LE_{smallest}) < 10^{10}$, which the user can check during initial explorations.

- Computing and storing LEs after every interval may not be necessary, especially if takes up a lot of memory. The field *Storage Steps* allows to control how often this is done.

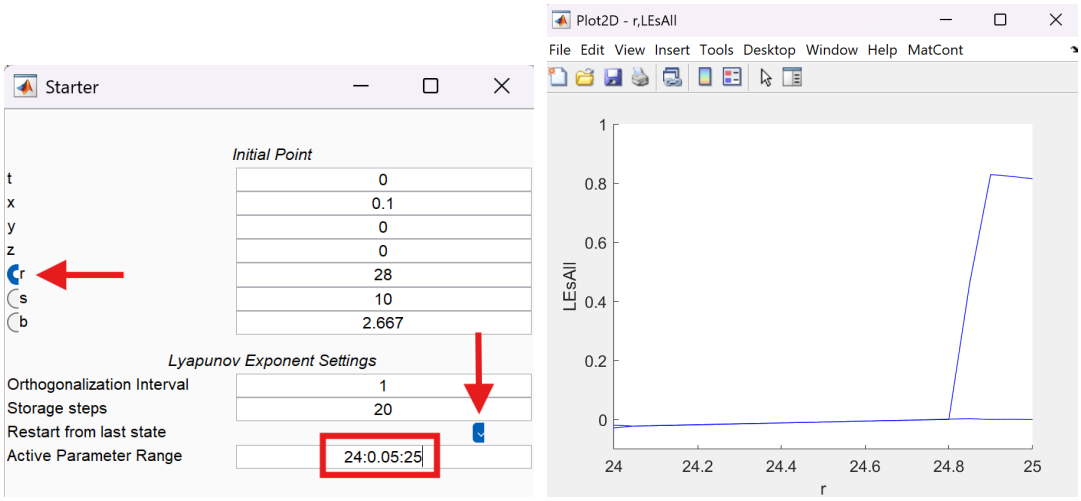


Figure 3: Left: Three changes in the settings so that we follow the attractor as r changes. Right: The two most positive Lyapunov exponents as the parameter r varies.

- If the box for *Restart from last state* is checked, the end state of the current computation is the new initial condition for the next computation. This way one can follow how an attractor evolves, which is useful if the outcome depends on the initial point or saves transient time. Otherwise, the first initial condition is taken for all computations. This only applies if the parameter range has more than one element.

- The field *Interval* should be large enough so that the average is converging. Or one may want to increase this value for more accuracy.

2.2 Command Line Mode

The same can be done with the following commands for a single parameter:

```
MySystem=feval(@lorenz0); % To create access to function handles
param=num2cell([28 10 8/3]); % To define parameter values correctly
x0=[.1;0;0]; % initial condition
[T,Y,TExp,Lexp,tf,yf]=LE(MySystem, [0 1000],x0,[],param,@ode45,1,1);
figure;plot(TExp,Lexp) % Running estimates of the Lyapunov Exponents
```

The final estimate of all exponents is given by `Lexp(end,:)`. You can add a loop to vary over a parameter, and use the last state `x0=Y(end,:)` for the next run.

References

[1] New features of the software MatCont for bifurcation analysis of dynamical systems. Dhooge, A. and Govaerts, W. and Kuznetsov, Yu.A. and Meijer, H.G.E. and Sautois B., Mathematical and Computer Modelling of Dynamical Systems, Vol. 14, No. 2, pp 147-175, 2008, DOI:https://doi.org/10.1080/13873950701742754.

[2] Calculation Lyapunov Exponents for ODE. Govorukhin, V. Available at https://nl.mathworks.com/matlabcentral/fileexchange/4628-calculation-lyapunov-exponents-for-ode